

Preliminary Study of Portable Vibrotactile and Electrical Stimulation System for Knee Proprioception Enhancement

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Abstract—Wearable robots assist elderly people with daily living activities like walking. The majority of walking or falling problems in the elderly stem from the reduction of skeletal muscle mass and proprioceptive performance. The existing wearable robots can help with muscle strength but not proprioception. In this paper, we perform a preliminary study of a lightweight, compact, and portable stimulation device for enhancing knee proprioception that combines stochastic resonance-based vibrotactile stimulation with transcutaneous electrical nerve stimulation. To assess proprioceptive acuity, we develop an experimental setup and protocol for a joint position matching test. Pilot tests of the proposed system with five young and healthy subjects are performed, and their results show a smaller matching error where vibrotactile and electrical stimulation is applied compared to where stimulation is not applied. We foresee that the developed system can be easily integrated into wearable robots to enhance muscle strength and proprioceptive function simultaneously.

I. INTRODUCTION

Along with the world population aging, the number of elderly people who have difficulty walking or with daily living activities is increasing. According to the United Nations [1], the global population aged 65 years or above is projected to rise from 10% in 2022 to 16% in 2050. Elderly people with a walking speed below 0.8 m/s, a clinically relevant threshold for gait speed chosen by several international sarcopenia working groups, are more likely to have poor physical performance in their health [2]. Elderly people with unstable walking may also fall down, leading to serious injury. Thus, the walking ability can be utilized to describe the extent of aging and has been considered crucial in improving their quality of life [3].

Lower limb wearable robotic devices have been researched to assist elderly people or patients in enhancing their walking ability or daily living activities [4], [5]. These are more advantageous than traditional therapeutic devices because they can tackle the limitations of traditional ones, such as high labor intensity and low repeatability [6]. The majority of walking or falling issues in elderly people stem from

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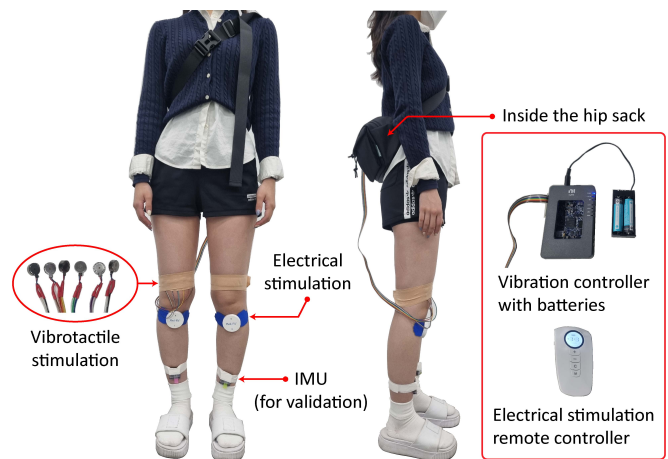


Fig. 1. The proposed portable and wearable vibrotactile and electrical stimulation system for knee proprioception enhancement

the loss of skeletal muscle mass and reduced function of the somatosensory system, especially proprioception. Since proprioception is the sense that lets us perceive the position, movement, and actions of parts of our body [7], it plays a crucial role in daily living activities as well as walking. However, the existing lower limb wearable robotic devices can assist only their reduced muscle strength but not their reduced proprioceptive function. Hence, it would be preferred to develop a wearable robotic device that can improve the proprioception of the operator.

In this paper, we conduct a preliminary study of stochastic resonance (SR)-based vibrotactile stimulation and transcutaneous electrical nerve stimulation (TENS) system for knee proprioception enhancement (See Fig. 1). SR-based vibrotactile stimulation is implemented with six linear resonance actuators whose amplitude and frequency are independently controllable. TENS is implemented with two self-adhering electrode patches. To assess proprioceptive acuity, we build a test bench setup and experimental protocol for a joint position matching test. Pilot tests of the developed system with five young and healthy subjects are carried out, and their results are evaluated and analyzed in detail. Since the proposed system is lightweight, compact, and portable, it is capable of seamless integration with the existing power-assisting exoskeletons.

The remainder of this paper is structured as follows. Section II explains the methods of vibrotactile stimulation and electrical stimulation. Section III describes the experimental

setup and protocol for the pilot test and evaluates its results by descriptive and inferential statistics. Section IV discusses the experimental results of pilot tests in terms of target position and learning effect. Finally, section V concludes this paper.

II. METHOD

A. Vibrotactile Stimulation

Stochastic resonance (SR) is a phenomenon where adding a certain level of noise improves signal detection [8]. SR-based vibrotactile stimulation has been shown to enhance proprioceptive acuity in postural stability in the elderly [9], [10], patients with Parkinson's disease [11], and stroke patients [12]. Subsensory mechanical noise is exerted on the position where the signal is stimulated or where an agonistic muscle is located, such as quadriceps femoris muscle [13]. Vibrotactile actuators can be classified by the working principle: eccentric rotating mass (ERM), linear resonance actuator (LRA), and piezoelectric actuator (PA). ERM uses spinning unbalanced mass in a magnetic motor to create vibrations but has a slow response and coupled amplitude and frequency. LRA pushes a mass up and down using magnetic coils and springs to create vibrations in a resonance frequency. PA utilizes the piezoelectric effect where a material is squeezed or stretched when an electrical signal is applied. LRA and PA are advantageous in that they have a fast response and can independently control their frequency and amplitude. However, since PA has low durability, so it is vulnerable to external disturbance, we consider LRA as a suitable option for the proposed portable and wearable stimulation system.

As shown in Fig. 1, we develop a vibrotactile stimulation device combining six LRAs (VG0832012, Vybronic, USA) with a pressure bandage, and it is attached to the quadriceps femoris muscle to stimulate the femoral nerve for knee proprioception enhancement. Each LRA has a diameter of 8 mm, a thickness of 3.25 mm, and a resonant frequency of 235 Hz. The frequency and amplitude of vibration are independently adjusted by a microcontroller (myRIO-1900, National Instruments, USA) with a programming language (LabVIEW, National Instruments, USA).

B. Electrical Stimulation

Electrical stimulation has been widely applied as a therapeutic treatment in a hospital. It can be classified by the frequency and amplitude of electrical signals: transcutaneous electrical nerve stimulation (TENS), interferential current therapy (ICT), and functional electrical stimulation (FES). TENS uses a low frequency of 1-200 Hz to stimulate the peripheral sensory nerve to relieve the body pain of patients based on gate control theory. ICT and FES use a relatively high frequency of 3,000-6,000 Hz and stimulate the motor nerve, which minimizes skin resistance compared to TENS. ICT utilizes interference of two or more currents with different frequencies. Since ICT requires bulky vacuum electrode cups for interferential current, it might not be suitable for the proposed portable and wearable stimulation system. FES is used to generate body movements artificially

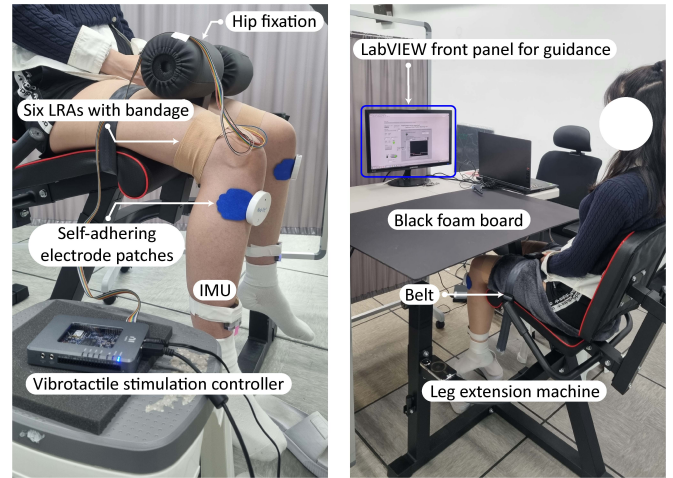


Fig. 2. Experimental setup for ipsilateral joint position matching test to assess proprioceptive acuity

in people with paraplegia due to injury to the central nervous system, so it is challenging to be applied to the elderly for proprioception enhancement. Thus, TENS is an appropriate option to improve proprioceptive acuity since it stimulates the peripheral sensory nerve and uses compact electrode patches easily attached to body parts.

TENS has been shown to improve proprioceptive acuity of postural balance and gait in chronic stroke patients [14]–[16]. TENS-based non-invasive and somatotopic sensory feedback approach can be employed to induce referred sensations to the phantom hand of amputees [17]. TENS can provide biofeedback and warning to humans to enhance seated balance [18]. In this paper, we utilize a wireless two-channel TENS machine (Med-Fit, USA) to stimulate the tibial nerve for knee proprioception enhancement, as shown in Fig. 1. Two self-adhesive conductive patches are attached to a knee, each with a size of 6×5 cm. Its waveform consists of asymmetric biphasic pulses, and its amplitude can be controlled in 20 stages. Since the proposed portable and wearable system consists of myRIO (193 grams), battery (117 grams), TENS remote controller (40 grams), and hip sack (250 grams), its total weight is 600 grams to be suitable for the integration to power-assisting exoskeletons.

III. PILOT TEST

A. Experimental Setup

The experimental setup of the proposed vibrotactile and electrical stimulation system for knee proprioception enhancement is shown in Fig. 2. A vibrotactile stimulation device is developed using six LRAs and stimulates the femoral nerve. Its frequency and amplitude can be simultaneously adjusted by the myRIO-1900. A wireless dual-channel rechargeable TENS machine stimulates the tibial nerve, and its amplitude is adjusted by the remote controller. To evaluate the proprioception enhancement performance of the proposed system, we conducted joint position matching tests to guess the joint angle only with proprioception in the

TABLE I
PROPRIOCEPTION ASSESSMENT PROTOCOL

Trial No.	Target Knee	Stimulation	
		Electrical (TENS)	Vibrotactile (SR)
1	Right	None	None
2	Left	None	None
3	Right	TENS	None
4	Left	None	None
5	Right	TENS	None
6	Left	None	None
7	Right	TENS	None
8	Left	None	None
Five-minute break			
9	Right	None	None
10	Left	None	None
11	Right	None	SR
12	Left	None	None
13	Right	None	SR
14	Left	None	None
15	Right	None	SR
16	Left	None	None
Five-minute break			
17	Right	None	None
18	Left	None	None
19	Right	TENS	SR
20	Left	None	None
21	Right	TENS	SR
22	Left	None	None
23	Right	TENS	SR
24	Left	None	None

absence of vision and auditory information. Since the knee joint is crucial for walking, we utilized knee joint angle for joint position matching tests. A leg extension machine and belt are also used to minimize other movements of the hip, waist, and ankle. An inertial measurement unit (IMU-9DOF, E2BOX, Republic of Korea) is attached to the bottom part of the tibia to measure the knee angle. A black foam board blocks the subject's vision not to utilize visual information during joint position matching tests.

Five young and healthy subjects (two male and three female; the age of 27 ± 4.2 years) volunteered in this experiment. None of the volunteers suffered from any musculoskeletal or neurological disorders. All subjects were given a practice session of the joint position matching test before performing the main experiment sets.

B. Experimental Protocol

The experimental protocol for knee proprioception assessment is based on the active ipsilateral joint position matching test [19]. 0 deg is calibrated as the knee joint angle in the most comfortable posture of each subject. Five target positions are -10 deg, 10 deg, 30 deg, 50 deg, and 70 deg. A total of 24 trials are performed, alternating right and left legs, as listed in Table I. Each subject is given a one-minute break between each trial and a five-minute break between every eight trials. Each trial has the following six procedures guided by the programmed LabVIEW front panel, including target position, knee joint angle, and holding time.

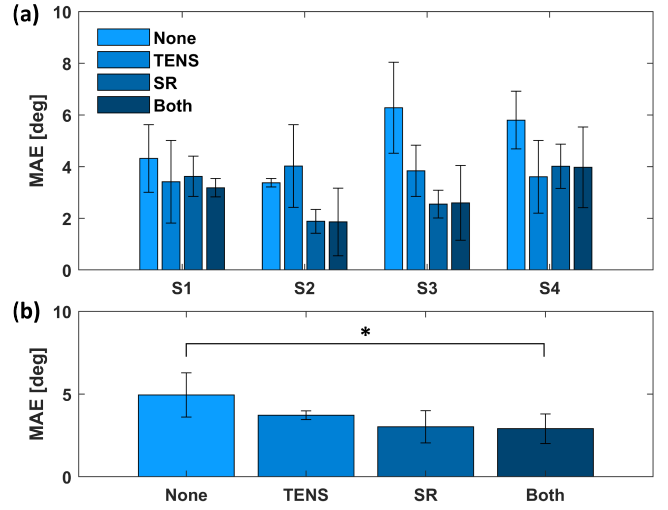


Fig. 3. Mean absolute error (MAE) and standard deviation of the right leg by stimulation group: (a) for each subject; (b) for all subjects (* $p < 0.05$).

TABLE II
MAE AND STANDARD DEVIATION OF THE RIGHT LEG (UNIT: DEG)

Subject	None	TENS	SR	Both
1	4.31 ± 1.31	3.41 ± 1.60	3.62 ± 0.78	3.18 ± 0.35
2	3.38 ± 0.16	4.02 ± 1.60	1.88 ± 0.46	1.86 ± 1.31
3	6.27 ± 1.76	3.83 ± 0.99	2.55 ± 0.54	2.60 ± 1.45
4	5.80 ± 1.11	3.60 ± 1.41	4.01 ± 0.86	3.97 ± 1.56
Total	4.94 ± 1.34	3.72 ± 0.26	3.02 ± 0.98	2.90 ± 0.90

- 1) Shuffle the sequence of five target positions and define the first target position as x deg.
- 2) Move the knee joint to 0 deg, seeing the joint angle. After that, hold the knee joint for 3 seconds.
- 3) Move the knee joint to x deg, seeing the joint angle. After that, hold the knee joint for 3 seconds.
- 4) Repeat 2).
- 5) Move the knee joint to x deg, **not** seeing the joint angle. After that, hold the knee joint for 3 seconds.
- 6) Change x deg to the next target position and repeat 2)–5) until five target positions are used.

The amplitude of TENS and SR was set to 90% of the threshold each subject could feel. The frequency of SR was set to 235 Hz, the same as the resonant frequency of LRAs. Experimental results of the right leg show whether vibrotactile and electrical stimulation can enhance proprioceptive sense or not. Experimental results of the left leg exhibit whether the proposed experimental protocol can be learned by repetition or not. The experimental protocol was approved by the Institutional Review Board of the Korea Advanced Institute of Science and Technology (KAIST) (KH2021-042).

C. Experimental Result

We analyzed the experimental results of the right leg for each subject (i.e., odd-numbered trials), as shown in Fig. 3. Descriptive statistics such as mean absolute error

TABLE III
WITHIN-SUBJECTS EFFECTS

Source	Method	DoF	F-value	p-value
Stimulation	Sphericity assumed	3	5.453	0.021
	Greenhouse-Geisser	2.012	5.453	0.044
	Huynh-Feldt	3	5.453	0.021
	Lower-bound	1	5.453	0.102
Error	Sphericity assumed	9		
	Greenhouse-Geisser	6.037		
	Huynh-Feldt	9		
	Lower-bound	3		

TABLE IV
PAIRWISE COMPARISONS

Stimulation		Mean difference	Standard error	p-value
None	TENS	1.223	0.708	0.183
	SR	1.923	0.644	0.058
	Both	2.039	0.564	0.036
TENS	None	-1.223	0.708	0.183
	SR	0.700	0.610	0.334
	Both	0.816	0.558	0.239
SR	None	-1.923	0.644	0.058
	TENS	-0.700	0.610	0.334
	Both	0.116	0.110	0.371
Both	None	-2.039	0.564	0.036
	TENS	-0.816	0.558	0.239
	SR	-0.116	0.110	0.371

(MAE) and its standard deviation are also listed in Table II. For each subject, each bar indicates the MAE of three trials of one stimulation group among none (trials 1, 9, 17), TENS (trials 3, 5, 7), SR (trials 11, 13, 15), and both (trials 19, 21, 23). We excluded experimental data from one subject by considering it as an outlier because its results were too far from those of other subjects. These data show that none group has higher error than the other stimulation group, meaning that vibrotactile and electrical stimulation can improve proprioceptive acuity.

To evaluate the experimental results of all subjects using inferential statistics, a repeated measures analysis of variance (ANOVA) test is performed by the statistical analysis program (SPSS Statistics 27, IBM, USA). Repeated measures ANOVA test requires the following three assumptions: normality, sphericity, and homogeneity of variance. The normality of experimental data was checked by the Shapiro-Wilk test, indicating p-values of 0.627 (none), 0.927 (TENS), 0.664 (SR), and 0.996 (both). The sphericity of experimental data was checked by Mauchly's sphericity test, showing an approximated chi-square $\chi^2(5) = 5.848$ and a p-value of 0.388. Homogeneity of variance (or homoscedasticity) can be assumed owing to the same number of trials for each subject. Since the sphericity of experimental data was already checked, repeated measures ANOVA test in assuming sphericity brought about $F(3,9) = 5.453$ and $p = 0.021$, meaning that it rejects the null hypothesis, as explained in Table III. In other words, the difference between the four stimulation groups is statistically significant. All pair-

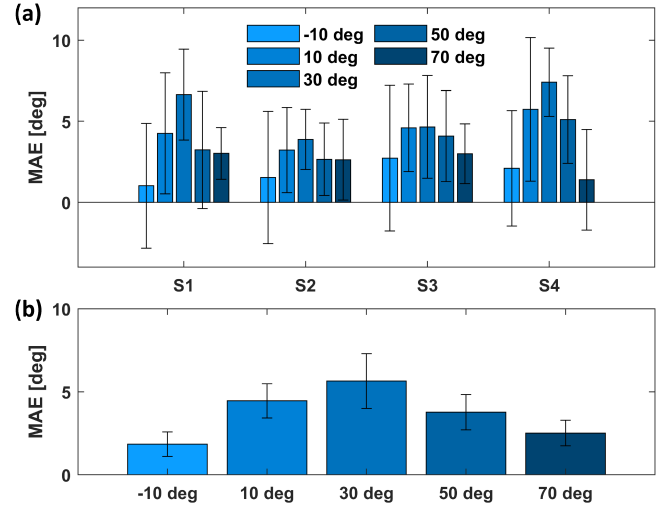


Fig. 4. MAE and standard deviation of the right leg by target position: (a) for each subject; (b) for all subjects.

wise comparisons in repeated measures ANOVA test are conducted using Fisher's least significant difference (LSD), and the results of statistical analysis are described in Table IV. These results show a significant difference of $p < 0.05$ between none group and both group.

IV. DISCUSSION

In this section, we discuss (i) whether target positions affect the matching error or not and (ii) the existence of a learning effect (i.e., the more experiments, the less matching error). In the ipsilateral joint position matching test, each trial includes five target positions of -10 deg, 10 deg, 30 deg, 50 deg, and 70 deg. Fig. 4 illustrates that errors in the joint position matching test are varied by target position. For each subject, each bar indicates the average of odd-numbered trials (right leg) of one target position. Experimental data indicate that 30 deg has a higher error and -10 deg and 70 deg have smaller errors than other positions. Legs are fully stretched or bent at 70 deg and -10 deg, so it can help proprioception to be activated. However, all subjects have difficulty moving their legs to 30 deg, which is a relatively ambiguous position. Mean and standard deviation are as follows: 1.84 ± 0.73 deg for -10 deg, 4.45 ± 1.03 deg for 10 deg, 5.64 ± 1.65 deg for 30 deg, 3.77 ± 1.07 deg for 50 deg, and 2.51 ± 0.77 deg for 70 deg. Fig. 5 shows the average matching errors of even-numbered trials (left leg) over time. It exhibits no increasing or decreasing trend of matching errors over time for each subject, which implies that the proposed experimental protocol has no learning effect.

V. CONCLUSION

This paper presented a portable and wearable vibrotactile and electrical stimulation system to enhance knee proprioception and its pilot test results. Vibrotactile stimulation is based on SR theory, and electrical stimulation is based on TENS. To validate the efficacy of the proposed stimulation

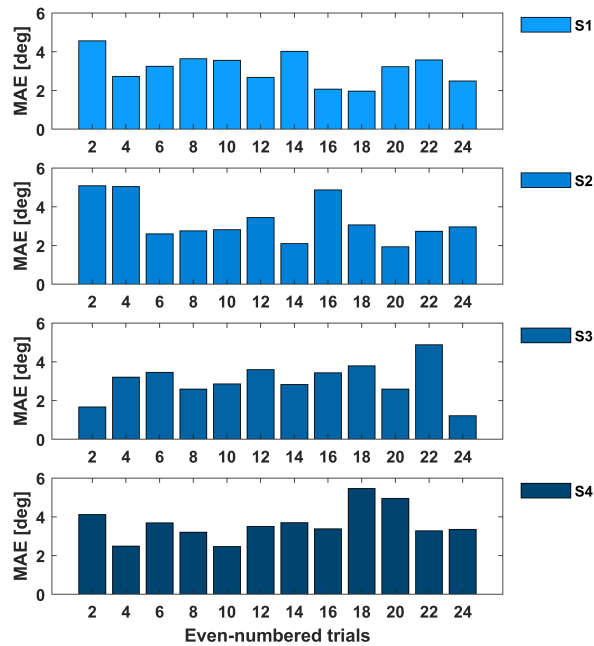


Fig. 5. MAE of the left leg by even-numbered trials for each subject to check learning effects

system, we build an experimental setup and protocol for an ipsilateral joint position matching test and conduct pilot tests with five young and healthy subjects. Experimental results are analyzed by repeated measures ANOVA tests, showing a significant difference of $p < 0.05$ between none group and both group. Our future work is to explore the proposed stimulation system by increasing the number of subjects and scrutinizing their experimental results. We hope that the proposed stimulation system is combined with the existing power-assisting wearable robotic devices to assist proprioceptive function as well as muscle strength for elderly people.

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